Fast Hyperspectral Image Classification Combining Transformers and SimAM-Based CNNs

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***Abstract*— Convolutional neural networks (CNNs) have been widely employed for hyperspectral image (HSI) classification due to their powerful ability to extract local spatial features. However, CNN-based methods cannot establish long-range dependencies among sequences of pixels. Transformers offer significant advantages when processing sequential data and can establish global relationships, but they still encounter a number of challenges, such as their limited spatial feature extraction ability, or their high computational cost. In order to address the aforementioned issues, we develop a new fast HSI classification approach combining transformers and SimAM-based CNNs. The latter is utilized to extract better spatial features, where the complex spatial characteristics of HSIs are retrieved using an improved hierarchical 2-D dense network structure. A dual attention unit (DAU) mechanism is then utilized to direct the model's attention to discriminative spatial pixel characteristics and effective feature map channels while suppressing information that is irrelevant for classification purposes. Regarding the spectral features, after extracting hierarchical local characteristics from various convolutional layers (using the hierarchical dense network structure), a squeezed-enhanced axial transformer is employed to establish global long-range dependencies while enhancing the ability of the model to extract local detail features in the HSI. Besides, a new Lion optimizer is utilized to improve the classification performance of our model. Our quantitative and comparative experiments on four benchmark datasets demonstrate that the effectiveness of the proposed approach provides better classification results than other state-of-the-art approaches. Moreover, our FTSCN also achieves better classification results than other methods in practical scenarios.**

***Index Terms*— Convolutional neural networks (CNNs), dual attention unit (DAU) mechanism, hyperspectral image (HSI) classification, squeezed-enhanced axial transformer, transformers.**

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I. Introduction

HYPERSPECTRAL image (HSI) classification is the process of categorizing each pixel in the HSI into one of several predefined classes based on spectral and spatial characteristics. HSIs are high-resolution images that comprise hundreds of narrow spectral bands, enabling the identification and discrimination of various materials and objects in the scene [1], [2]. With the development of remote sensing imaging technology, HSIs play an important role in current remote sensing studies and are widely applied in many important applications, such as environmental monitoring [3], land use planning [4], and target detection [5], among others. Classification, as an important exploitation technique, has become an active research topic in remote sensing image processing [6], [7], [8]. Numerous machine learning methods have been adopted for HSI classification purposes, including support vector machines (SVMs) [9], attribute profiles [10], morphological profiles [11], random forests (RFs) [12], or K-nearest neighbor (KNN) [13]. However, the abovementioned traditional feature extraction approaches (which are usually based on handcrafted characteristics) belong to the category of shallow methods; namely, the information contained in the HSI is not fully utilized. Thus, the shallow features provided by these methods often fail to capture the deep and complex structure of HSIs, resulting from a weak feature extraction ability that makes them less robust than other methods in complex situations [1], [7].

With the rapid advancement of deep learning (DL) in computer vision, many new robust techniques for HSI processing have emerged [6], [7], [14], [15], [16]. As a result, many well-known DL approaches have been successfully implemented for HSI categorization purposes, such as convolutional neural networks (CNNs) [17], [18], [19], deep belief networks (DBNs) [20], [21], recurrent neural networks (RNNs) [22], stacked autoencoder (SAE) [23], generative adversarial networks (GANs) [24], and the graph convolutional network (GCN) [25]. Chen et al. [17] pioneered the use of DL for HSI classification utilizing the SAE for obtaining joint

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spectral–spatial features. Likewise, a DBN-based framework was proposed for HSI classification in [20], demonstrating great potential. Zhong et al. [21] enhanced the representation ability of DBNs for HSI classification by pretraining and fine-tuning the procedure via regularization of diversity-promoting prior potential factors. Zhu et al. [24] adopted GANs to effectively alleviate overfitting in CNNs, significantly improving HSI classification accuracy. Zhang et al. [25] first introduced GCN into cross-scene HSI classification and developed characterize nonlocal topological structure relationships and local semantic information transfer networks by combining GCN and CNN.

RNNs are very efficient in the task of handling data with sequential features and are also able to extract semantic feature information for long sequences of features. Since HSI spectral bands are essentially feature vectors with sequential arrangement, RNNs offer significant potential to become a powerful spectral signature extraction tool for HSI classification purposes [22], [26], [27], [28]. Mou et al. [22] suggested using the RNN model to handle spectral band sequences to characterize the correlation among such HSI bands. Wu et al. [26] adopted a bidirectional convolutional long short-term memory (LSTM) network for mining the relationship between adjacent spectral bands in HSIs. Zhang et al. [27] proposed using RNNs to capture semantic characteristics from local spatial sequences, as well as a nonlocal spatial sequence method that allows the RNN model to integrate information from nonlocal—but similar—samples. Ma et al. [28] proposed a new framework for HSI categorization based on an LSTM that decreases the model's computational complexity by constructing a sequence of features from the segmented target pixel. It should be noted that, even though RNN- or LSTM-based methods can effectively represent spectral sequence features, they can hardly utilize the spatial–contextual information contained in HSIs.

Since CNNs exhibit outstanding local contextual modeling ability, they have been extensively applied for spectral–spatial HSI classification. Wang et al. [29] proposed a fast dense spectral–spatial convolution model (FDSSC) that improved the propagation of feature information between various spectral–spatial convolution layers by combining several dense blocks. The hierarchical residual network with attention mechanism (HRAM) approach was adopted to boost the utilization of hierarchical characteristic information while allocating appropriate weights to various spatial and spectral characteristics using the attention mechanism in [30]. Hang et al. [31] embedded attention-assisted modules in each convolutional layer of the CNN, allowing the model to focus on salient spatial information or channel features. Alkhatib et al. [32] proposed a three-branch 3-D-CNN model that employs three types of convolutional kernels to extract HSI features in each of the three networks, providing robust spectral, spatial, and spectral–spatial characteristics. To fully utilize the spectral–spatial features in HSIs, Liang et al. [33] employed RNN-based models to retrieve spectral sequential features, while a CNN-based multiscale dense network was used to mine complex spatial–contextual characteristic information. Although the CNN-based methods above exhibit advantages in terms of spatial–contextual characterization, they still face important limitations when processing spectral sequential data (and CNNs often fail to efficiently depict long-range dependencies).

The visual transformer (ViT) has recently demonstrated significant promise in computer vision [34]. By combining a multiheaded self-attentive (MHSA) module—describing global remote dependencies in the input sequential data—and a multilayer perceptron (MLP), the transformer (and its variants) has achieved equivalent or superior performance than CNN-based approaches in tasks such as image processing, target recognition, and anomaly detection [34], [35]. The transformer architecture, which is distinct from that of CNNs, is a self-attention framework combined with an MLP capable of modeling the long-range dependence of tokens in sequences [36], [37]. As such, transformers are increasingly being employed for HSI classification purposes [38], [39], [40], [41], [42], [43], [44], [45], [46], [47]. Based on the MHSA mechanism, He et al. [38] pioneered the use of bidirectional encoder representation from the transformer (BERT) for HSI classification, which flexibly captures the global dependencies between pixels utilizing the global receptive fields in BERT. Hong et al. [39] utilized the transformer to learn spectral sequence information from adjacent bands of HSIs, using groupwise spectral embedding to learn locally detailed representations of the spectral sequence features, and a cross-layer adaptive fusion module to reduce information loss in layer-by-layer propagation. He et al. [40] proposed an approach that integrates CNNs with transformers to compensate for the inability of CNNs to capture long sequence feature relationships. To efficiently obtain deep spatial feature information and establish dependencies among spectral sequences, Peng et al. [41] proposed a spatial–spectral transformer with a cross-attention mechanism. Zou et al. [43] developed a local-enhanced spectral–spatial transformer that consists primarily of two modules: the first one transforms the HSI into self-adaptive spectral–spatial tokens, while the second one (a local-enhanced transformer encoder) is used to reinforce the local features while preserving the information from remote features. Yang et al. [44] employed ViTs and convolution operations to classify HSIs for the first time, embedding convolution operations in the transformer architecture to obtain minor spectrum variances and express local spatial background information. Liu et al. [45] designed a scaled dot-product central attention to obtain the spectral–spatial characteristic information of the query pixel and the pixels similar to the central pixel on the HSI patches. Simultaneously, a central attention network is presented by merging the characteristics of the hidden layer customized by HSI with the dense connectivity of the spectral signature information of the query pixels. Based on the CNN and the transformer architecture, a spectral–spatial feature tokenization transformer (SSFTT) was proposed in [46], which converted the low-level convolutional spectral–spatial features captured by the CNN into semantic tokens and then fed them to the transformer

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encoder for feature representation. Song et al. [47] proposed a combination of the 3-D-CNN and a bottleneck spatial–spectral transformer (BS2T) for HSI categorization, employing a multiheaded spatial–spectral self-attention mechanism to build long-range global dependencies among HSI pixels to exploit the local–global feature information contained in the image. However, this approach suffers from a relatively large amount of computations and long training time.

In order to alleviate the problems of the abovementioned methods, we propose a fast HSI classification approach that integrates the transformer and the SimAM-based CNN architecture. The SimAM is a complete 3-D weighted, parameter-free attention mechanism that can effectively improve the feature extraction ability of CNNs [48]. Our approach, called FTSCN, first improves the 2-D-CNN using SimAM and then extracts the complex spatial feature information contained in HSIs using a hierarchical 2-D dense network structure. Then, a dual attention unit (DAU)—with both spatial and channel attention mechanisms—is utilized to allocate larger weight parameters to discriminative spatial pixels and feature maps associated with important feature channels in the model while suppressing the information that contributes little to the final classification outcomes. Regarding the spectral features, after extracting layered local features from various convolutional layers using a hierarchical 3-D dense network structure, we use a squeezed-enhanced axial transformer (with a detail enhancement kernel) to establish global long-range dependencies while improving the extraction of local detail characteristics. In addition, to boost the categorization performance of our FTSCN model, a new Lion optimizer is employed for the first time for HSI processing instead of the traditional Adam optimizer.

Overall, our new FTSCN model presents several innovative contributions that can be summarized as follows.

1) The CNN architecture is improved using a SimAM attention mechanism to improve the feature extraction ability of CNNs. The rich spatial information contained in HSIs (and stored in-between multiple convolutional layers) is retrieved using an improved hierarchical 2-D dense network structure. The DAU mechanism is then utilized to focus the model on pixels with discriminative spatial information and effective feature maps.
2) Regarding the spectral features, the feature extraction capability of our model is enhanced using a squeezed-enhanced axial transformer, where the global long-range dependencies are accurately characterized, while the extraction of local detail features is enhanced using a convolution-based detail enhancement kernel. To the best of our knowledge, this is the first time in the literature that a squeezed enhanced axial transformer has been adopted for HSI categorization.
3) To improve the model's classification performance, a new Lion optimizer is utilized for the first time for HSI classification. Furthermore, an extensive comparison using four benchmark HSI datasets demonstrates the effectiveness of the proposed FTSCN approach, which is able to significantly reduce the model training time.

The remainder of this article is structured as follows. Section II briefly introduces some related works. Section III describes the architectural details of our FTSCN method. Section IV provides the experimental setup, results, and a detailed comparison to state-of-the-art methods, followed by a discussion. This article is concluded with some comments and hinted at plausible future research in Section VI.

II. Related Work

A. SimAM Attention Mechanism

SimAM, a complete 3-D, weighted, and parameter-free attention mechanism, was proposed based on mature neuroscientific theories in [48]. In comparison to existing channel attention and spatial attention mechanisms, SimAM produces a full 3-D weighted attention module where both mechanisms coexist and jointly facilitate information selection during visual processing, achieving good performance in terms of both speed and accuracy. The phenomenon that the most informative neurons in visual neuroscience exhibit various firing patterns in the surrounding neurons and maintain the activity of the surrounding neurons is called spatial suppression [49]. Active neurons with a significant spatial suppressive effect on the activity of surrounding neurons usually have a greater impact on the outcome of the visual processing task. These neurons can be identified by calculating the linear separability of a target neuron from other neurons. Moreover, the edge characteristics of the image are identical to those of the spatially suppressed neurons, which usually present incredibly high contrast with regard to the surrounding texture characteristics. Thus, by employing an energy function to allocate 3-D weights to various visual tasks without additional parameters, the SimAM attention mechanism can efficiently improve the characteristic extraction capability of CNNs, as shown in Fig. 1. Here, considering that the energy function treats each pixel of the HSI feature map as a neuron, and the minimum energy of the neuron can be denoted as [48]

$$e_t = \frac{\frac{4}{M}\sum_{i=1}^{M}(x_i-\alpha)^2 + 4\beta}{(t-\alpha)^2 + \frac{2}{M}\sum_{i=1}^{M}(x_i-\alpha)^2 + 2\beta}$$
$$\alpha = \frac{1}{M}\sum_{i=1}^{M} x_i \tag{1}$$

where t and i denote the target neuron of the input feature $\boldsymbol{X} \in R^{H\times W\times C}$ and the index of the spatial dimension, respectively. H, W, and C correspondingly denote that the HSI data containing $H \times W$ pixels and C spectral bands. M stands for the number of neurons on the channel, with $M = H \times W$, and β is a parameter value that is typically taken as $1e-4$. α denotes the average value calculated for all neurons in the channel (except for the target neuron t), and x_i denotes other neurons (except the target neuron t) within the same channel of the input feature $\boldsymbol{X}$.

Since spatially suppressed neurons exhibit high linear separability (somewhat different from other neurons), they suffer from obvious deviations both in α and t, resulting in lower values of e_t. Furthermore, (1) demonstrates that the lower the energy of the neuron, the more distinguishable the

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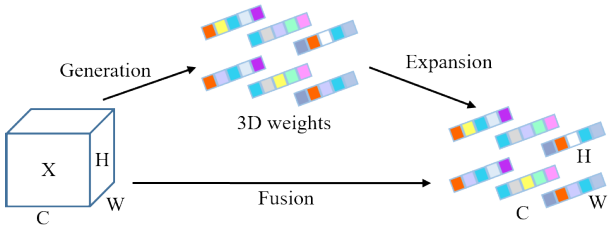


Fig. 1. Graphical illustration of the SimAM attention mechanism, where the complete 3-D weights are for attention.

neuron t is from the other neurons around it. Thus, the value of the weight parameter for each neuron can be calculated by $(1/e_t)$. The feature matrix is then augmented in accordance with the definition of the attention mechanism, and the formula is calculated as

$$\bar{\boldsymbol{X}} = \text{sigmoid}\left(\frac{1}{\boldsymbol{E}}\right) \odot \boldsymbol{X} \tag{2}$$

where $\boldsymbol{E}$ denotes the entire set of weights $(1/e_t)$ grouped in channel and spatial dimensions. $\bar{\boldsymbol{X}}$ denotes the output feature map, which has the same feature map size as input $\boldsymbol{X}$. Finally, the sigmoid activation function is employed for determining the weight parameters of each neuron in each location.

B. Transformer

CNNs are relatively efficient in conducting local feature extraction, but they are not excellent at capturing long-range dependencies. The ViT has a strong global field of perception, but it is inefficient in computing self-attention. Many researchers have tried to develop efficient implementations of ViT [50], [51], [52], [53], [54]. Zhang et al. [50] proposed a TopFormer that applies global attention to the original input data, but this resulted in the loss of information and subsequently poor semantic segmentation performance. Based on the original transformer, Xie et al. [51] aggregated information from various layers by utilizing an MLP decoder, combining local and global information to achieve powerful modeling capability. Wan et al. [54] proposed a mobile-oriented plug-and-play squeeze-enhanced axial transformer module that achieves an optimal balance between accuracy and inference efficiency. Here, a global attention mechanism is employed for the feature map $\boldsymbol{x} \in R^{H\times W\times C}$ and can be formulated as

$$\boldsymbol{y}_o = \sum_{p\in G(o)} \text{softmax}_p\left(\boldsymbol{q}_o^T \boldsymbol{k}_p\right)\boldsymbol{v}_p \tag{3}$$

where $\boldsymbol{q}$, $\boldsymbol{k}$, and $\boldsymbol{v}$ denote $\boldsymbol{W_q x}$, $\boldsymbol{W_k x}$, and $\boldsymbol{W_v x}$, respectively. $\boldsymbol{W_q x}$, $\boldsymbol{W_k x} \in R^{C_{qk}\times C}$, and $\boldsymbol{W_v x} \in R^{C_v\times C}$ denote the learnable weight matrices. $G(o)$ refers to all locations on the feature map for position $o = (i, j)$. The above attention module suffers from a relatively complexity $O(W^2H^2(C_v + C_{qk}))$, resulting in an inefficient model.

Luong et al. [52] developed a window-based attention approach to enhance the effectiveness of the global attention model, which can be represented as

$$\boldsymbol{y}_o = \sum_{p\in N_{m\times m}(o)} \text{softmax}_p\left(\boldsymbol{q}_o^T \boldsymbol{k}_p\right)\boldsymbol{v}_p \tag{4}$$

where $N_{m\times m}(o)$ denotes the $m \times m$ neighborhood around o. While this strategy reduces the model's complexity to $O(m^2WH(C_v + C_{qk}))$, it is subject to the loss of global information because of the use of a fixed window. Ho et al. [53] proposed an axial attention module that can be expressed as

$$\boldsymbol{y}_o = \sum_{p\in N_{1\times W(o)}} \text{softmax}_p\left(\boldsymbol{q}_o^T \boldsymbol{k}_p\right)\boldsymbol{v}_p + \sum_{p\in N_{H\times 1(o)}} \text{softmax}_p\left(\boldsymbol{q}_o^T \boldsymbol{k}_p\right)\boldsymbol{v}_p \tag{5}$$

where $N_{H\times 1}(o)$ and $N_{1\times W}(o)$ denote all the positions of column o and row o, respectively. The complexity is reduced to $O((W + H)WH(C_v + C_{qk}))$, but this self-attention mechanism leads to the loss of global dependence information. To overcome this shortcoming, a concise squeezed axial attention module is added to aggregate global feature information via the extraction of global semantic features. Besides, a convolution-based detail enhancement kernel is adopted to supplement local details, obtaining an efficient squeezed enhanced axial attention method [54]. Using the squeeze operation, global information is retained by a single axis (horizontal and vertical), greatly lowering the computational complexity of the global semantic extraction process.

III. Proposed Method

Fig. 2 provides a flowchart of the proposed FTSCN model, which consists of three major components: a spatial branch, a spectral branch, and a feature fusion branch network. In the spatial branch, the SimAM attention mechanism is employed to assign proper weight parameters to spatial information neurons and suppress irrelevant information from surrounding neurons, improving the 2-D-CNN architecture. Afterward, a hierarchical dense network structure is built using our SimAM-based 2-D-CNN to retrieve the hierarchical information contained in the original HSI. After that, a DAU mechanism with spatial and channel attention properties is utilized to improve the capability of the model to acquire spatial characteristics. In the spectral branch, a 3-D-CNN is utilized to build a hierarchical dense network structure to extract hierarchical local characteristics from each convolutional layer. Then, a squeeze-enhanced axial transformer (combined with a detail enhancement kernel) is adopted to boost the spectral signature extraction capacity of the model. Finally, the two branching network features are combined by a feature fusion module using fully connected (FC) layers to generate a new joint spatial–spectral signature that is used for classification purposes. In the following, we provide a detailed explanation of the individual components of our method.

A. Hierarchical Dense Network for Feature Extraction

It is known that different convolutional layers in CNNs extract distinct types of feature information. In shallow convolutional networks, the model extracts specific shallow features. As the number of convolutional layers increases, the perceptual field increases, allowing the model to extract deeper

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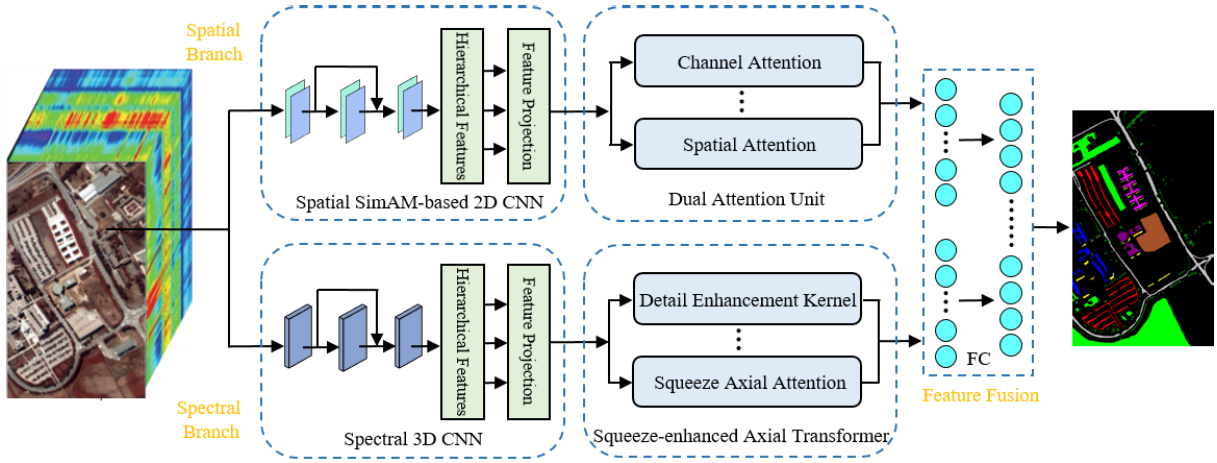


Fig. 2. Flowchart of the proposed FTSCN model that contains a spatial branch, a spectral branch, and a feature fusion branch network. The SimAM-based 2-D-CNN is exploited to build a hierarchical dense network structure to retrieve the hierarchical spatial feature information contained in the original HSI. After that, DAU with spatial and channel attention properties is utilized to improve the capability of the model to acquire spatial characteristics. Besides, the 3-D-CNN is utilized to build a hierarchical dense network structure to extract hierarchical local spectral characteristics from each convolutional layer, and a squeeze-enhanced axial transformer is adopted to boost the spectral signature extraction capacity of the model.

semantic feature information. Hence, fully exploiting the hierarchical feature information of various convolutional layers can effectively boost the multiscale feature representation ability of the DL-based HSI classification approach. SimAM can provide appropriate weight parameters to spatially informative neurons in the CNN while suppressing the information coming from surrounding neurons with irrelevant information. As a result, we employ SimAM to improve the 2-D-CNN architecture in this article. Specifically, we build a hierarchical dense network structure using a SimAM-based 2-D-CNN to retrieve complex spatial information contained in the raw HSI. The architecture of our SimAM-based hierarchical dense network is illustrated in Fig. 3, where 3-D-Conv and 2-D-Conv stand for 3-D and 2-D convolution, respectively, BN stands for batch normalization, and mish refers to the mish activation function.

First, a convolutional kernel of size (1, 1, k_b) is employed to lower the spectral band dimensionality of the original input HSI data patch, and then, a hierarchical dense network (built using a SimAM-based 2-D-CNN) is employed for processing the reduced spatial characteristic map of the HSI to obtain multiscale hierarchical spatial characteristic information contained in various convolutional layers. This information is supplied to the DAU module to improve the model's capacity to extract spatial characteristics. In Fig. 3, (k_{sa}, k_{sa}) and m_i represent the size of the convolutional kernel and the number of feature channels acquired from each convolutional layer in the SimAM-based dense network, respectively. The value of i is an integer from 1 to 5.

To adequately utilize the hierarchical spectral features coming from various convolutional layers and effectively boost the ability to perform spectral signature extraction from the original HSI, a hierarchical 3-D dense network is employed for extracting spectral characteristics in the spectral branch, as illustrated in Fig. 2. Following that, a squeeze-enhanced axial transformer (combined with a detail enhancement kernel) is applied to establish the global long-range dependencies and boost the ability to perform signature extraction of local detail features. The hierarchical 3-D dense network contains five convolutional layers in total. More details regarding the experimental parameter setup for the spatial and spectral branches will be given in Section IV-D.

B. Dual Attention Unit

Traditional HSI classification models give equal weights to all spatial pixels and feature maps of channel features, respectively. As a matter of fact, different spatial pixels and feature maps contribute unequally to categorization outcomes. The attention mechanism, including spatial and channel attention modules, allows the model to concentrate more on discriminative spatial pixels and effective feature maps while suppressing information that is detrimental to classification. In addition, the DAU combines spatial attention [55] and channel attention [56] to share information inside the feature tensor, including spatial and channel dimensions, and allowing the model to assign higher weights to discriminative spatial pixels and feature maps [57]. Based on these two attention modules, Roy et al. [58] considerably improved the model's ability to represent features by utilizing channel attention to model the interrelationships across channels of convolutional feature maps, selectively stressing the significance of each channel, and suppressing invalid feature maps. Gao et al. [59] created a triple-branch attention module by combining the spatial and channel attention modules and introducing a spectral attention mechanism, which significantly enhances the characteristic extraction ability of the model. It is not only capable of determining which spatial regions of feature maps merit more attention via the spatial attention module but it is also capable of determining which feature maps are critical for learning via the channel attention module and reallocating weights to feature maps of different channels. Inspired by this, we employ DAU in the spatial branch to boost the extraction of characteristic spatial information. The structure of the DAU is illustrated graphically in Fig. 4. It includes two modules: channel attention and spatial attention.

Let us assume that the input feature map is given by $\boldsymbol{M} \in R^{H_0 \times W_0 \times C_0}$, where $H_0 \times W_0$ denotes the spatial size

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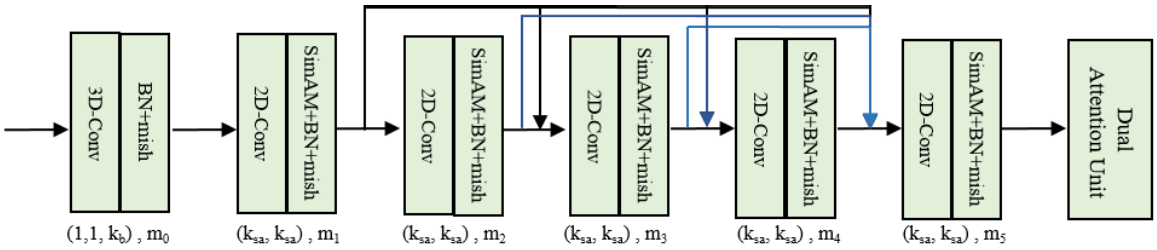


Fig. 3. Architecture of our SimAM-based hierarchical dense network, where 3-D-Conv and 2-D-Conv stand for 3-D and 2-D convolutions, respectively. BN and mish stand for batch normalization and the activation function, separately.

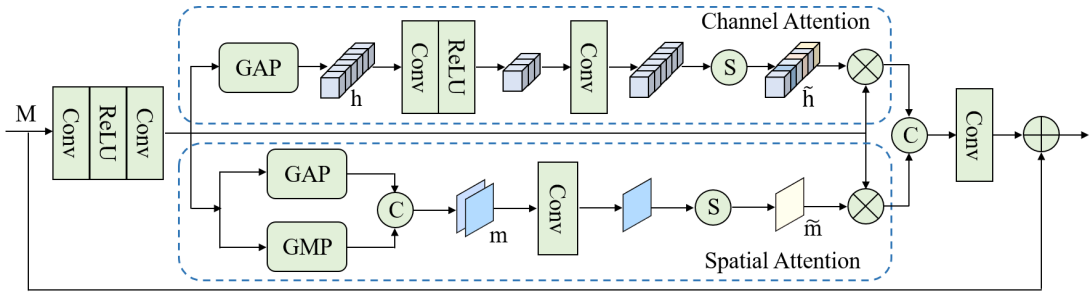


Fig. 4. Structure of the DAU, which includes spatial and channel attention mechanisms. GAP and GMP stand for global average and maximum pooling operations, respectively. Ⓒ denotes the concatenation operation. Ⓢ denotes the sigmoid function. ⊕ and ⊗ stand for element-level summation and multiplication, separately.

and C_0 denotes the number of channels in the feature maps. First, the input feature map $\boldsymbol{M}$ is processed by two 3×3 convolutional layers to produce the new feature map $\boldsymbol{M_0}$, which is then fed into the spatial and channel attention modules, respectively. Channel attention establishes relationships among feature channels of the convolutional feature maps by employing squeeze and excitation operations. The squeeze operation encodes the global context via global average pooling across spatial dimensions, so as to generate a new feature map $\boldsymbol{h} \in R^{1\times1\times C_0}$. After processing $\boldsymbol{h}$ using two convolutional layers in the excitation operation, a new attentional feature map $\tilde{\boldsymbol{h}}$ is obtained by feeding a sigmoid activation layer with a self-gating mechanism. Finally, the feature map $\boldsymbol{M_0}$ is reweighted by multiplying $\tilde{\boldsymbol{h}}$ by the feature map $\boldsymbol{M_0}$ to generate a new feature output $\boldsymbol{M_1}$.

For spatial attention, the global average pooling and maximum pooling operations are employed to aggregate spatial feature information (in pixelwise fashion) to produce feature vectors $\boldsymbol{M}^c_{\text{avg}} \in R^{H_0\times W_0\times1}$ and $\boldsymbol{M}^c_{\max} \in R^{H_0\times W_0\times1}$, which have the same size and number of feature maps. The new feature map $\boldsymbol{m} \in R^{H_0\times W_0\times2}$ is then produced by concatenating them. The map $\boldsymbol{m}$ is processed by the convolution and sigmoid activation layers to get the spatial attention map $\tilde{\boldsymbol{m}} \in R^{H_0\times W_0\times1}$, which is calculated as

$$\tilde{\boldsymbol{m}} = \delta(\text{Conv}(\text{Concat}(\text{GAP}(\boldsymbol{M}), \text{GMP}(\boldsymbol{M})))) \tag{6}$$

where δ stands for the sigmoid activation function and Conv denotes a convolutional layer. GAP and GMP stand for global average and maximum pooling layers, respectively. Finally, similar to the channel attention module operation, the feature map $\boldsymbol{M_0}$ is reweighted to generate a new feature $\boldsymbol{M_2}$. Subsequently, the output $\boldsymbol{M_1}$ of the channel attention module and the output $\boldsymbol{M_2}$ of the spatial attention module are processed by the connection operation and then fed to the convolutional layer to obtain new features, which assign higher weight values to both discriminative spatial pixels and channels with important feature maps.

C. Squeeze-Enhanced Axial Transformer

In order to aggregate the global characteristic information contained in the original HSI while reducing the computational complexity of the global semantic information extraction process, a squeeze-enhanced axial attention transformer is employed to establish the long-range dependencies on the spectral features of the HSI for enhancing the extraction of spectral characteristics. A schematic of the squeeze-enhanced axial transformer module is given in Fig. 5. It is composed primarily of a squeezed axial attention module for extracting global semantic features and a convolution-based detail enhancement kernel module for enhancing local details [54].

In the squeeze-enhanced axial attention transformer, the global information of the HSI is retained on a single axis (horizontal and vertical) using the squeeze operation, which greatly reduces the computational complexity of the global semantic extraction process. Specifically, averaging the feature maps in the horizontal and vertical directions, respectively, to achieve extrusion in each direction is accomplished as follows:

$$\begin{aligned} \boldsymbol{q}(h) &= \frac{1}{W}(\boldsymbol{q}^{\rightarrow(C_{qk},H,W)_{I_W}})(H, c_{qk}) \\ \boldsymbol{q}(v) &= \frac{1}{H}(\boldsymbol{q}^{\rightarrow(C_{qk},H,W)_{I_H}})(W, c_{qk}) \end{aligned} \tag{7}$$

where $\boldsymbol{q}^{\rightarrow(\cdot)}$ denotes a permutation of the dimensions of the given tensor $\boldsymbol{q}$ and $I_m \in R^m$ represents a vector in which all elements are equal to 1. The squeeze axial attention expression

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obtained above is given by

$$\boldsymbol{y}_o = \sum_{p=1}^{H} \text{softmax}_p\left(\boldsymbol{q}_{(h)i}^T \boldsymbol{k}_{(h)p}\right)\boldsymbol{v}_{(h)p} + \sum_{p=1}^{W} \text{softmax}_p\left(\boldsymbol{q}_{(v)j}^T \boldsymbol{k}_{(v)p}\right)\boldsymbol{v}_{(v)p} \quad (8)$$

where $\boldsymbol{q}_{(h)}, \boldsymbol{k}_{(h)}, \boldsymbol{v}_{(h)} \in R^{H\times C_{qk}}$ and $\boldsymbol{q}_{(v)}, \boldsymbol{k}_{(v)}, \boldsymbol{v}_{(v)} \in R^{W\times C_{qk}}$. Each location of the feature map propagates information only about the axial features that are squeezed horizontally and vertically. However, since the aforementioned process does not include the feature map's position awareness, an extrusion axial position embedding was used to squeeze the axial attention to obtain the positional-aware squeeze axial attention, which can be formulated as

$$\boldsymbol{y}_{(i,j)} = \sum_{p=1}^{H} \text{softmax}_p\left(\left(\boldsymbol{q}_{(h)i} + \boldsymbol{r}_{(h)i}^q\right)^T\left(\boldsymbol{k}_{(h)p} + \boldsymbol{r}_{(h)p}^k\right)\right)\boldsymbol{v}_{(h)p} + \sum_{p=1}^{W} \text{softmax}_p\left(\left(\boldsymbol{q}_{(v)j} + \boldsymbol{r}_{(v)j}^q\right)^T\left(\boldsymbol{k}_{(v)p} + \boldsymbol{r}_{(v)p}^k\right)\right)\boldsymbol{v}_{(v)p} \quad (9)$$

where $\boldsymbol{r}_{(\theta)}^q, \boldsymbol{r}_{(\theta)}^k \in R^{H\times C_{qk}}$ denotes the horizontal or vertical position embedding, which enables $q_{(\theta)} and k_{(\theta)}$ to obtain its position in the axial feature. θ indicates the horizontal or vertical position, which takes the values of h and v. Although horizontal and vertical squeeze operations effectively capture the global semantic information of the HSI, spatially localized information is lost. Therefore, as shown in Fig. 5, an aided convolution-based kernel is employed to reinforce the spatial local detail information.

In the detail enhancement kernel module, first, $\boldsymbol{q}$, $\boldsymbol{k}$, and $\boldsymbol{v}$ are obtained from $\boldsymbol{x}$ and a set of learnable weight matrices, as shown in 3, and then transferred to a deep convolutional layer of size 3×3 size and to a BN layer for processing, leading to the aggregation of auxiliary local feature information from $\boldsymbol{q}$, $\boldsymbol{k}$, and $\boldsymbol{v}$. The dimensionality $2Cqk+Cv$ is compressed to C, and the detail enhancement weight matrix is generated after processing by the rectified linear unit (ReLU) activation function, a 1×1 convolutional layer, and a BN layer. Finally, the enhanced features are combined with the squeeze axial attention features by a multiplication operation.

D. Spectral–Spatial Feature Fusion

The spectral and spatial characteristics gained through the spectral and spatial branches are dimension-reduced using the GAP layer, and the new characteristics after the dimensionality reduction are processed by the FC layer and then connected to produce new joint spatial–spectral characteristics. Finally, the new spatial–spectral features obtained from the FC layer are transmitted into the softmax layer to achieve the final classification results.

In addition, to boost the categorization performance of the FTSCN model, a new Lion optimizer is utilized instead of the traditional Adam optimizer. Different from traditional artificial optimizers, such as Adam, Lion is a programmatic autodiscovery optimization algorithm [60]. Traditional manual optimizers must save first- and second-order moments, whereas Lion simply keeps track of momentum and computes updates for each parameter using sign function operations. Compared to Adam and traditional manual optimizers, the uniform update calculated using the sign function frequently produces a bigger norm, which allows Lion to require a smaller learning rate (LR) in network training.

IV. Experimental Results and Analysis

In this section, quantitative and comparative experiments are executed on four benchmark datasets to validate the performance of the proposed FTSCN approach with regard to other state-of-the-art approaches. First, we introduce the datasets. Then, we provide the parameter settings, experimental results, and a detailed discussion and analysis. Then, sample sensitivity testing is performed on the compared experimental approaches, and the computational times of different techniques are compared. Ablation experiments are also designed to verify the effectiveness of various modules in the model. Eventually, various methods are tested on a practical HSI data scenario to further verify the effectiveness of the FSTCN method.

A. Datasets' Description

Four benchmark HSI datasets have been used in our experiments.

1) The University of Pavia (UP) dataset was acquired by the reflective optics spectrographic imaging system (ROSIS) in Pavia, Italy, in 2002. The UP contains 610×340 pixels with a spatial resolution of 1.3 m/pixel and a spectral wavelength range of 430–860 nm. After removing 12 noisy bands, we retain 103 spectral bands and nine ground-truth classes comprising 42 776 pixels.
2) The Indian Pines (IP) dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) across Northwestern Indiana, USA. It has 145×145 pixels, a spatial resolution of 20 m/pixel, and a spectral band coverage range of 400–2500 nm. There are 220 spectral bands altogether, 20 of which are removed due to water absorption. This scene has 16 ground-truth classes comprising 10 249 pixels.
3) The Kennedy Space Center (KSC) dataset was collected by the AVIRIS sensor in Florida, USA. It has 512×614 pixels, a spatial resolution of 18 m/pixel, and a spectral band coverage range of 400–2500 nm. There are 176 spectral bands altogether after discarding some water absorption and noisy spectral bands. This scene has 13 ground-truth classes comprising 5211 pixels.
4) The Zaoyuan (ZY) dataset was acquired by the Operative Modular Airborne Imaging Spectrometer (OMIS) in ZY area, China. It has 137×202 pixels and a spectral band coverage range of 400–1700 nm. There are 80 spectral bands altogether after discarding some noisy spectral bands. This scene has eight ground-truth classes comprising 23 821 pixels.

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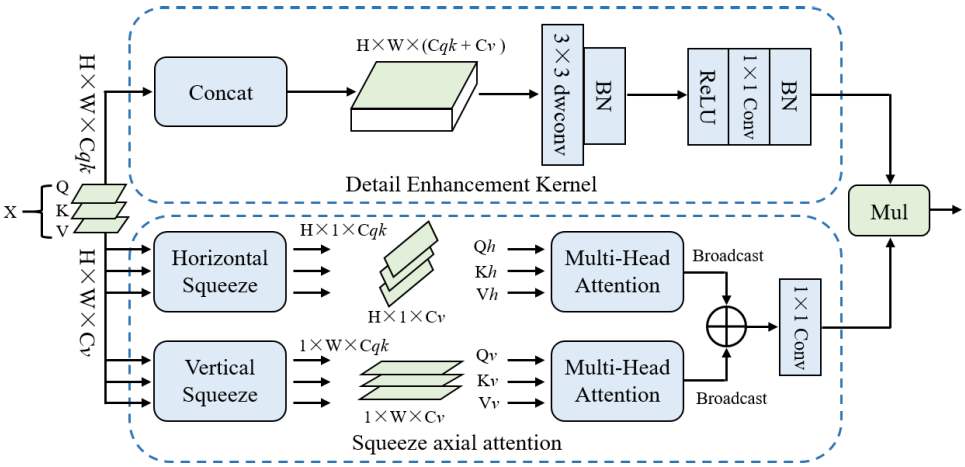


Fig. 5. Architecture of the squeeze-enhanced axial transformer, where ⊕ denotes the addition operation and Mul denotes the multiplication operation.

B. Experimental Configuration and Evaluation Indicators

To verify the validity of the FTSCN approach, we compared it with several advanced algorithms, among them SSRN [61], FDSSC [29], TriCNN [32], DBDA [62], HRAM [30], SSFTT [46], and BoS2T [47]. These methods are briefly summarized in the following.

1) The SSRN structure is built on ResNet and 3-D-CNN, with spectral and spatial signature extraction handled by a spectral ResNet block and a spatial ResNet block.
2) FDSSC is a fast and dense spectral–spatial convolution model, which extracts spectral and spatial hierarchical characteristic information using 3-D-CNN dense blocks with convolution kernel sizes of $1 \times 1 \times 7$ and $3 \times 3 \times 1$, respectively.
3) The TriCNN is a three-branch network framework built on the 3-D-CNN architecture, with each branch network composed of three convolutional kernels of various scales, $3 \times 3 \times 1$, $3 \times 3 \times 3$, and $1 \times 1 \times 3$, respectively. The feature information extracted from the three branches at various scales is concatenated and fused using the FC layer technique.
4) The DBDA approach is based on a dual-attention mechanism network composed of two dense 3-D-CNN blocks with convolutional kernel sizes of $1 \times 1 \times 7$ and $3 \times 3 \times 1$, respectively, while simultaneously employing spatial and channel attention mechanisms to improve feature acquisition.
5) The HRAM approach uses a two-branch structure of hierarchical residual networks to extract spatial and spectral information at various scales. Two attention mechanisms are also implemented to balance the weights of hierarchical spectral and spatial characteristics.
6) The SSFTT is a spectral–spatial feature tokenization transformer approach built on the CNN and transformer architecture, which converted the low-level convolutional spectral and spatial characteristics captured by 2-D and 3-D convolution layers into semantic tokens and then fed them to the transformer encoder for feature representation.
7) The BoS2T incorporates the 3-D-CNN with the BS2T, which extracts local spatial and spectral characteristic information individually using 3-D dense blocks and then constructs long-range global dependencies of HSI pixels using an MHSA module within the BS2T.

To objectively analyze the classification accuracy of the FTSCN model, three metrics were used, namely, overall accuracy (OA), average accuracy (AA), and kappa coefficient. Moreover, classification maps were also used for qualitative visual evaluation [63]. To ensure the robustness of the experiments, for UP, IP, KSC, and ZY, 0.4%, 4%, 2.5%, and 2.5% of the labeled samples were randomly picked as training and validation samples, with the remainder serving as the test set. Meanwhile, the classification scores are reported as the mean and variance of 15 Monte Carlo experiments.

All experiments were performed on the computer equipped with an Intel CPU i9-9900K, 128-GB RAM, and 2080 Ti GPU, and the experimental environment is PyTorch 1.7 and Python 3.7.1.

C. Parameter Settings

Parameter settings have an essential effect on how well the model performs in DL-based network model creation. Therefore, extensive experiments are carried out to analyze and identify those key parameters that determine the performance of the FTSCN model, including batch size, LR, spatial size, and convolutional kernel size. Besides, there are also certain model parameters that we determined empirically. The LR determines the gradient descent rate and the convergence of the FTSCN when training the model. When the LR of the model is too large, it may lead to periodic oscillations. In the opposite case, it may lead to very slow or even failed convergence. Empirically, the Lion optimizer's ideal LR is generally three to ten times lower than that of the Adam optimizer [60]. Thus, the LR of the FTSCN gets set as 0.0001 for optimal performance. The training periods of the FTSCN model are set to 150 for IN and ZY, and 200 for UP and KSC.

Larger batch sizes may necessitate more memory in order to retain the weight update parameters, whereas smaller ones may result in model nonconvergence and underfitting. The selection

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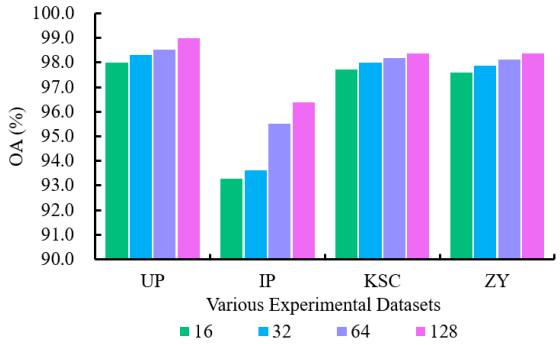


Fig. 6. Variation of classification accuracy for different batch sizes on various datasets.

of a proper batch size enhances the model's convergence speed and classification accuracy while also saving training time. The variation of classification accuracy for different batch sizes on various datasets is illustrated in Fig. 6. From this figure, the optimal batch size parameter is set to 128.

Generally, large convolutional kernels can provide a larger perception field to obtain more global characteristic information, but they will also introduce more interference information and reduce the model's categorization performance. The variation of classification performance with various sizes of spatial and spectral convolutional kernels is depicted in Fig. 7. From this figure, we can conclude that, in order to achieve optimal categorization performance, the values 5×5 and $1 \times 1 \times 7$ are employed to set the spatial and spectral convolution kernel sizes, respectively. Similarly, after conducting experiments and analyzing the influence of the input patch size range on the pixel-level categorization task, we determined the ideal input data size for spatial and spectral branches. Specifically, the spatial size of the patch for the spatial branch network in UP, IP, KSC, and ZY is set to 23×23, 17×17, 23×23, and 19×19, respectively, while the corresponding data cube sizes for the spectral branch network are $3 \times 3 \times 103$, $3 \times 3 \times 220$, $5 \times 5 \times 176$, and $5 \times 5 \times 80$.

In addition, the parameter configuration of the spatial and spectral branches in the FTSCN approach is shown in Table I. In the spatial branch, k_b in the convolution kernel $(1, 1, k_b)$ of the first convolution layer of the SimAM-based hierarchical 2-D dense network takes the value of the HSI's spectral band number, where the stride s_{sa0} gets set to 1, and the padding p_{sa0} gets set as 0. In the other four hierarchical convolution layers, k_{sa} in the convolution kernel (k_{sa}, k_{sa}) is set to 5. As for the spectral branch, the convolutional kernel size is $(1, 1, k_{se})$ in the previous four convolutional layers of the hierarchical 3-D dense network, where k_{se} gets set as 7. The stride s_{se0} and the padding p_{se0} are set to 2 and 0, separately. However, the stride s_{se} and the padding p_{se} take values of 1 and 3, respectively. In the last convolution layer, the size of k_{se0} in the convolution kernel $(1, 1, k_{se0})$ depends on k_{se}. When the value of spectral channel B of the HSI is odd, k_{se0} takes the value $\text{floor}((B - k_{se})/2 + 1)$. On the contrary, k_{se0} takes the value $\text{floor}((B - k_{se} + 1)/2)$. Here, floor(.) denotes rounding to the nearest whole number. k_{se0} in the convolution kernel $(1, 1, k_{se0})$ is 2 in the last convolution layer. The corresponding values of s_{sa} and p_{sa} are taken as 1 and 2. Besides, DAU_Attention() and Sea_Attention() signify the DAU attention mechanism module and the squeeze-enhanced axial transformer module, respectively, and their configuration parameters are chosen as the default optimal parameters.

In addition, an early stopping strategy is implemented in the training process to prevent the FTSCN model from overfitting and also to reduce the training time. That is, if the loss function does not lower after a given number of epochs (15 in the FTSCN model), we halt training.

D. Classification Results

To compare the performance of the FTSCN approach fairly with that of other methods, the optimal parameters stated for such methods in the reference publications were used in all experiments. Tables II–V show the classification results obtained by all the compared methods on UP, IP, KSC, and ZY, with the optimal outcomes are highlighted in bold. Meanwhile, Figs. 8–11 show the corresponding classification maps.

1) UP Dataset: According to Table II, the FTSCN approach provides 99.00% OA, 99.07% AA, and 98.68% kappa with the UP dataset, which greatly surpasses the other compared approaches. Specifically, in terms of OA, the FTSCN outperforms the method with the poorest accuracy (TriCNN) by 10.43 %, while it surpasses the second most accurate method (BoS2T) by 1.22%. The classification accuracy of TriCNN is only 88.57% in terms of OA, which is likely due to the fact that, while TriCNN employs a three-branch network with 3-D convolutional kernels at three different scales to gather feature information, and the oversimplified traditional CNN structure and small number of training samples fail to make the model capture enough distinguishable spectral and spatial characteristics. Remarkably, SSRN, HRAM, FDSSC, and DBDA outperform TriCNN in terms of OA by 2.94%, 6.86%, 5.55%, and 5.51%, respectively. This is because the ResNet-based SSRN and HRAM methods, as well as the dense 3-D-CNN-based DBDA and FDSSC, utilize a hierarchical structure that enables the transmission and exploitation of feature information between different convolutional layers. Although SSFTT, BoS2T, and FTSCN all integrate the benefits of CNNs and transformers (not only extracting local information with the CNN but also establishing the long-range global dependencies of HSI pixels with the transformer), the categorization accuracy of SSFTT in terms of OA is only 94.47%, which is 4.53% and 1.22% lower than the accuracy of BOS2T and FTSCN, respectively. This is due to the fact that SSFTT merely employs one 3-D convolution and one 2-D convolution layer to capture local spectral and spatial characteristics, and fails to capture deep semantic characteristics. In contrast, BoS2T and FTSCN adopt a hierarchical structure based on the dense network, which can appropriately extract the hierarchical semantic feature information of the HSI, which actually improves the model's feature recognition ability. In addition to that, the FTSCN method uses a squeeze-enhanced axial transformer to establish long-range global dependencies and boosts the model's ability to mine detailed information, while the improved 2-D-CNN

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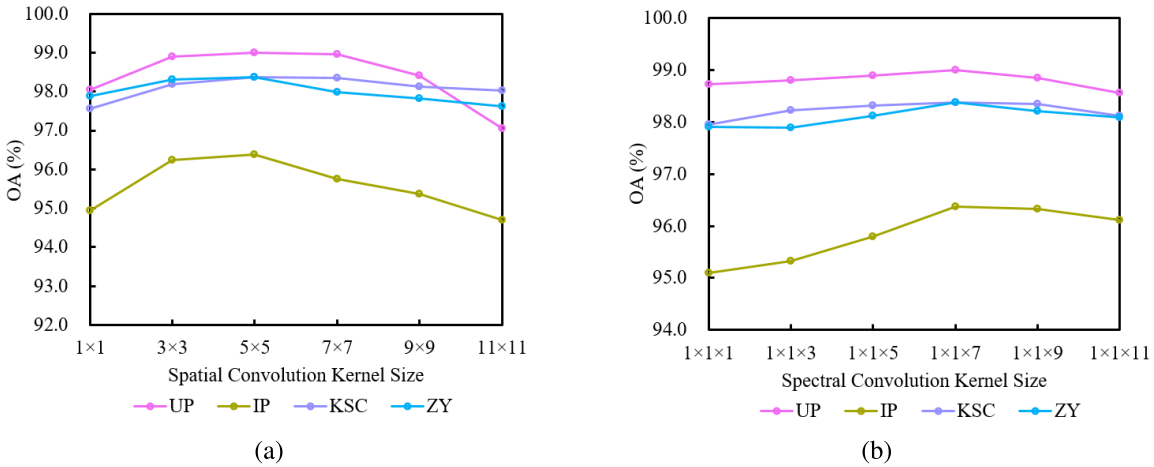


Fig. 7. Variation of FTSCN classification performance with various sizes of (a) spatial and (b) spectral convolutional kernels.

TABLE I
PARAMETER CONFIGURATION OF THE SPATIAL AND SPECTRAL BRANCHES IN THE FTSCN MODEL

Network	Layers	Convolution Kernel / Operation	Setup
Spatial Branch	3D-Conv_0	(1,1, k_b), 48	stride:s_{sa0}, padding:p_{sa0}, BN+mish
	2D-Conv_1	(k_{sa}, k_{sa}), 24	stride:s_{sa}, padding:p_{sa}, SimAM+BN+mish
	2D-Conv_2	(k_{sa}, k_{sa}), 12	
	Concat_1	concat(2D-Conv_1, 2D-Conv_2), 36	BN+mish
	2D-Conv_3	(k_{sa}, k_{sa}), 12	stride:s_{sa}, padding:p_{sa}, SimAM+BN+mish
	Concat_2	concat(2D-Conv_1, 2D-Conv_2, 2D-Conv_3), 48	BN+mish
	2D-Conv_4	(k_{sa}, k_{sa}), 12	stride:s_{sa}, padding:p_{sa}, SimAM+BN+mish
	Concat_3	concat(2D-Conv_1, 2D-Conv_2, 2D-Conv_3, 2D-Conv4), 60	BN+mish
	2D-Conv_5	(k_{sa}, k_{sa}), 60	stride:s_{sa}, padding:p_{sa}, SimAM+BN+mish
	DAU_Attention()		
Spectral Branch	3D-Conv_1	(1, 1, k_{se}), 24	stride:s_{se0}, padding:p_{se0}, BN+mish
	3D-Conv_2	(1, 1, k_{se}), 12	stride:s_{se}, padding:p_{se}, BN+mish
	Concat_1	concat(3D-Conv_1, 3D-Conv_2), 36	BN+mish
	3D-Conv_3	(1, 1, k_{se}), 12	stride:s_{se}, padding:p_{se}, BN+mish
	Concat_2	concat(3D-Conv_1, 3D-Conv_2, 3D-Conv_3), 48	BN+mish
	3D-Conv_4	(1, 1, k_{se}), 12	stride:s_{se}, padding:p_{se}, BN+mish
	Concat_3	concat(3D-Conv_1, 3D-Conv_2, 3D-Conv_3, 3D-Conv_4), 60	BN+mish
	3D-Conv_5	(1, 1, k_{se0}), 60	stride:s_{se}, padding:p_{se0}, BN+mish
	Sea_Attention()		

based on the SimAM attention module can assign more weights to spatially informative neurons and suppress the irrelevant information coming from surrounding neurons.

According to the classification maps in Fig. 8, the TriCNN (with the lowest classification accuracy) clearly results in a substantial number of misclassified pixels for the UP dataset. The ResNet-based SSRN and HRAM outperformed TriCNN in terms of classification significantly, but there were significant misclassifications in the "Gravel" and "Bare soil" classes. Although the classification maps provided by FDSSC and DBDA—based on a dense network hierarchical structure—outperform all previous approaches in terms of visual appearance, they have misclassified pixels in the "Bare soil" and "Bricks" categories and are not as homogeneous as those provided by BoS2T and FTSCN. This is due to the fact that BoS2T and FTSCN not only use dense networks to extract hierarchical feature information, but they also use a transformer to determine the long-term dependence between pixels. In addition, FTSCN appears to be clearly superior to BoS2T in terms of the provided classification maps, particularly in the "Asphalt" and "Bricks" classes, indicating that FTSCN outperforms BoS2T in terms of detail information augmentation.

2) IP Dataset: The IP dataset has a relative imbalance among sample categories. As demonstrated in Table III, the categories with a small number of samples have only one or two training samples in the experiment, but the category with a larger number of samples reaches 99. However, FTSCN still attained the highest classification accuracy in contrast to other approaches, with OA of 96.38%, AA of 97.34%, and kappa of 95.87%. TriCNN provided the worst performance in terms of OA, with 88.76%, whereas SSRN, HRAM, FDSSC, and DBDA approaches are 2.94%, 6.86%, 5.55%, and 5.51% better than TriCNN, correspondingly. This demonstrates that the use of convolutional layers with a hierarchical structure can substantially facilitate the exploitation of the hierarchical features provided by convolutional layers and boost the model's accuracy in classification. Although SSFTT and BoS2T obtained classification accuracies of 95.72% and 95.55% in terms of OA, their accuracies for AA were 4.6% and 6.14% lower than that of FTSCN, respectively. This also demonstrates that FTSCN, which combines a dense

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TABLE II
CLASSIFICATION ACCURACY OF VARIOUS APPROACHES WITH THE UP DATASET

Class	SSRN	FDSSC	TriCNN	DBDA	HRAM	SSFTT	BoS2T	FTSCN
C1 (27/6631)	96.58	97.24	87.51	97.10	94.57	95.26	98.19	**98.43**
C2 (75/18649)	97.07	98.58	98.42	99.03	98.02	**99.96**	99.12	99.62
C3 (9/2099)	85.24	90.64	64.12	89.23	94.21	75.51	93.89	**100.00**
C4 (13/3064)	99.65	99.82	82.50	99.12	97.28	86.50	99.45	**99.95**
C5 (6/1345)	99.91	99.71	99.99	99.82	97.65	**100.00**	99.36	**100.00**
C6 (21/5029)	92.80	98.23	86.31	97.41	97.83	93.41	98.30	**99.55**
C7 (6/1330)	81.09	89.54	48.85	86.86	92.21	99.92	95.62	**99.74**
C8 (15/3682)	83.58	85.69	74.85	88.34	80.50	82.17	91.49	**94.38**
C9 (4/947)	97.23	99.28	80.78	98.47	97.84	86.53	95.88	**100.00**
OA/%	93.77 (1.20)	96.06 (0.75)	88.57 (2.10)	96.32 (0.55)	95.06 (0.95)	94.47 (1.36)	97.78 (0.19)	**99.00 (0.21)**
AA/%	92.57 (1.42)	95.41 (0.92)	80.37 (4.28)	95.05 (1.12)	94.46 (1.14)	91.03 (2.15)	96.81 (0.42)	**99.07 (0.23)**
Kappa/%	91.74 (1.95)	94.78 (0.81)	84.69 (3.11)	95.14 (0.72)	93.43 (1.31)	92.62 (2.33)	97.05 (0.25)	**98.68 (0.28)**

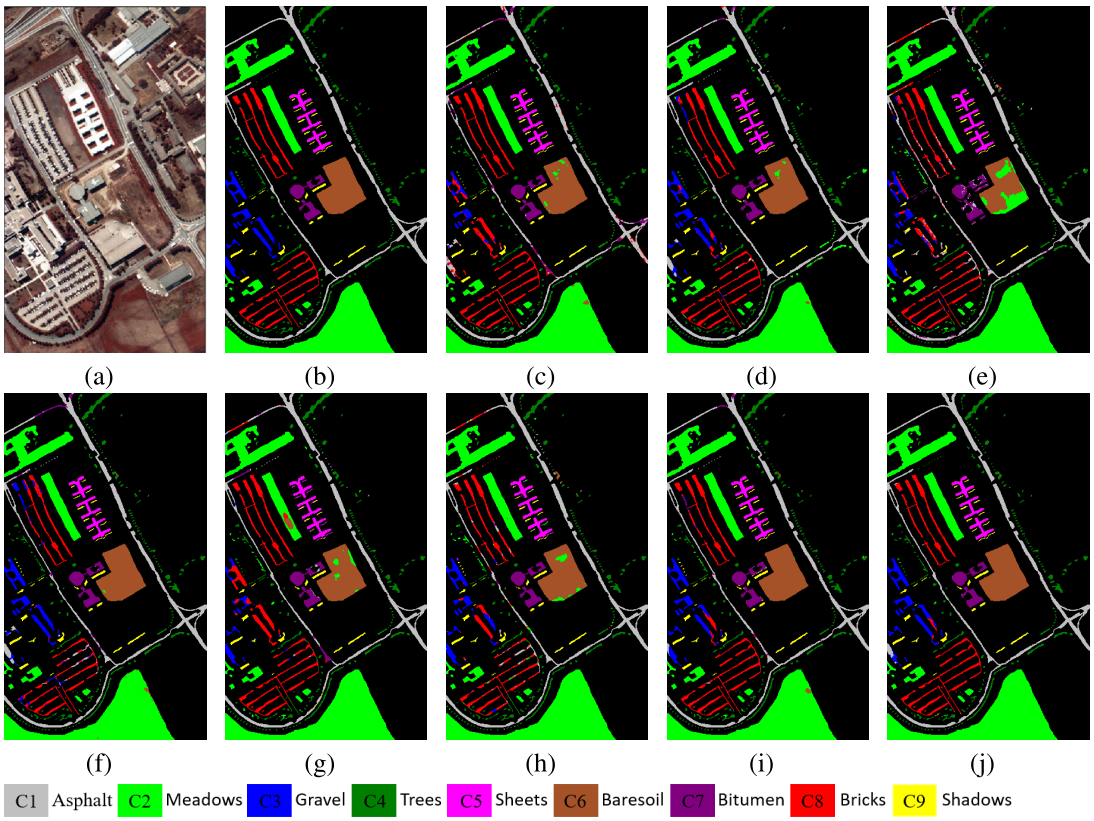


Fig. 8. Classification visual maps obtained by various approaches with the UP dataset. (a) False-color map. (b) Ground truth. (c) SSRN (93.77%). (d) FDSSC (96.06%). (e) TriCNN (88.57%). (f) DBDA (96.32%). (g) HRAM (95.06%). (h) SSFTT (94.47%). (i) BoS2T (97.78%). (j) FTSCN (99.00%).

network hierarchical structure with a squeeze-enhanced axial transformer, is not only effective in acquiring HSI feature information but also offers considerable advantages when dealing with the sample imbalance problem. Furthermore, the classification maps in Fig. 9 show that SSFTT and FTSCN have a relatively fewer number of misclassifications and noise than other approaches. Nonetheless, FTSCN has fewer misclassified pixels on "Corn-mintill," "Grass-trees," and "Soybean-clean" categories than SSFTT and is relatively better at preserving border detail information and structure.

3) KSC Dataset: As shown in Table IV, TriCNN and SSFTT have the lowest classification accuracy among all methods tested with the KSC dataset, with OAs of 63.70% and 72.65%, respectively. This is due to the fact that the number of training examples in this experiment is insufficient (class C13 with the most samples has just 24 samples), preventing the model from learning the semantic characteristics of HSIs sufficiently. Although DBDA and HRAM achieve classification accuracies of 95.19% and 95.16% in terms of OA, there are relatively more misclassified pixels, and the classification visual maps in Fig. 10 appear to be too smooth. The FTSCN provided the highest classification accuracy with 98.37% for OA, 97.56% for AA, and 98.18% for kappa, which exceeds the classification accuracy of the equivalent indexes of

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TABLE III
CLASSIFICATION ACCURACY OF VARIOUS APPROACHES WITH THE IP DATASET

Class	SSRN	FDSSC	TriCNN	DBDA	HRAM	SSFTT	BoS2T	FTSCN
C1 (2/46)	60.00	86.82	54.55	**98.72**	95.80	86.36	97.79	94.80
C2 (58/1428)	89.88	**93.86**	88.36	90.45	93.85	92.63	91.26	90.11
C3 (34/830)	95.11	90.66	81.77	90.04	94.00	95.23	**95.55**	94.38
C4 (10/237)	88.33	95.95	56.80	94.13	94.63	96.05	96.47	**97.25**
C5 (20/483)	98.61	96.83	68.32	98.48	98.11	99.57	**100.00**	99.90
C6 (30/730)	96.25	98.08	97.46	97.93	98.00	98.29	98.46	**99.43**
C7 (2/28)	65.06	63.74	33.33	77.20	69.73	**100.00**	64.32	98.52
C8 (20/478)	95.45	96.80	98.59	98.65	99.96	**100.00**	99.91	99.24
C9 (1/20)	12.73	50.00	23.16	70.00	93.21	63.16	49.32	**100.00**
C10 (39/972)	90.49	91.01	87.22	93.81	93.49	91.64	94.89	**96.86**
C11 (99/2455)	87.62	94.59	94.77	96.43	97.16	98.94	96.32	**97.31**
C12 (24/593)	90.44	93.83	82.63	90.83	90.03	84.01	96.40	**97.74**
C13 (9/205)	99.48	97.28	96.62	96.90	98.65	99.99	96.47	**100.00**
C14 (51/1265)	96.73	98.59	97.69	98.84	98.09	**99.01**	97.89	97.80
C15 (16/386)	91.03	95.71	76.89	93.01	**95.70**	92.45	92.33	95.57
C16 (4/93)	94.48	93.79	81.36	93.70	93.57	86.52	91.85	**98.57**
OA/%	91.70 (1.25)	94.31 (1.51)	88.76 (2.09)	94.27 (1.09)	95.62 (0.76)	95.72 (0.67)	95.55 (0.34)	**96.38 (0.14)**
AA/%	84.48 (4.05)	89.85 (5.29)	76.22 (4.78)	92.44 (0.42)	94.00 (0.93)	92.74 (1.39)	91.20 (0.48)	**97.34 (0.31)**
Kappa/%	90.50 (1.45)	93.50 (2.43)	87.12 (2.31)	93.49 (0.95)	95.00 (0.81)	95.12 (0.70)	94.93 (0.38)	**95.87 (0.16)**

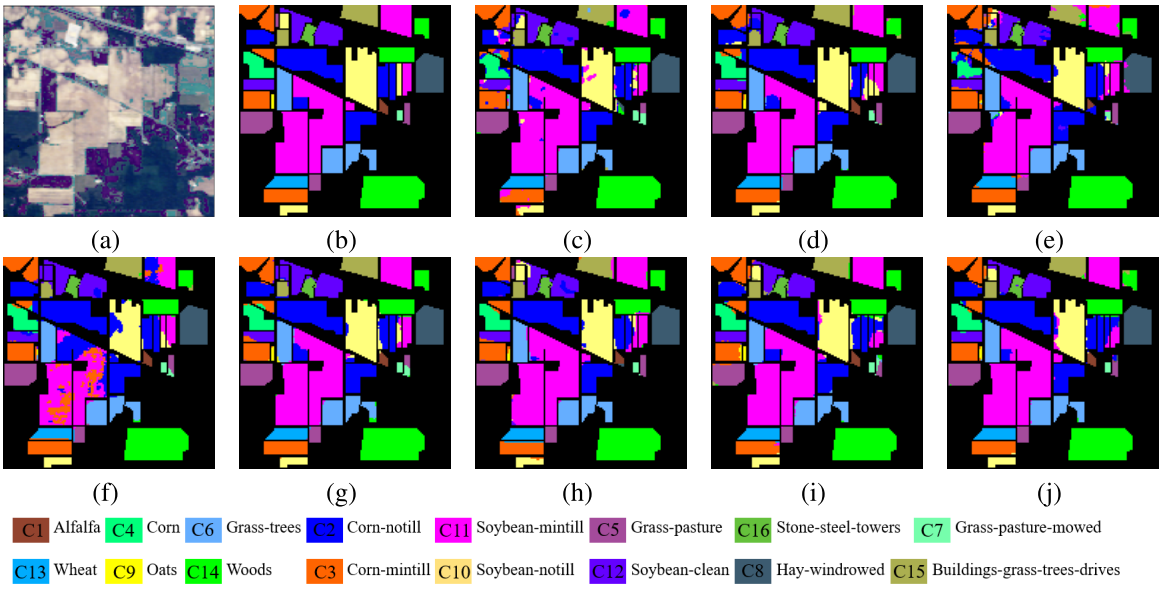


Fig. 9. Classification visual maps obtained by various approaches with the IP dataset. (a) False-color map. (b) Ground truth. (c) SSRN (91.70%). (d) FDSSC (94.31%). (e) TriCNN (88.76%). (f) DBDA (94.27%). (g) HRAM (95.62%). (h) SSFTT (95.72%). (i) BoS2T (95.55%). (j) FTSCN (96.38%).

BoS2T by 0.55%, 1.63%, and 0.59%, respectively. Meanwhile, Fig. 10 demonstrates that the classification map of FTSCN has considerably less classification noise than that of BoS2T, and FTSCN can extract more detailed features, which results in better spatial homogeneity. In addition, the categorization accuracies of all compared approaches are unsatisfactory on the "slash pine" class, with accuracies below 92.28%. The cause may be that the intraclass variability of the "slash pine" class is high, and the model captures little prior information due to the limited number of training samples available, resulting in poor generalization performance.

4) ZY Dataset: From Table V, it can be seen that the FTSCN method yields the optimal classification accuracy (in terms of OA, AA, and kappa) for the ZY dataset, which are 98.37%, 97.71%, and 97.86%, respectively. In comparison, BoS2T achieves an accuracy of 98.10% in terms of OA, which is 0.27% lower than that of FSCTN. This demonstrates that the use of a squeeze-enhanced axial transformer is capable of establishing long-term dependencies and boosting the acquisition of detailed information. Moreover, the improved 2-D-CNN with the SimAM attention module (and the utilization of spatial and channel attention) assigns greater weights to important spatial information neurons, which can efficiently boost the spatial characteristic extraction capability of the model. As demonstrated in Fig. 11, the classification maps provided by FTSCN, BoS2T, and DBDA not only have less noisy and misclassified pixels than those provided by SSRN, FDSSC, TriCNN, HRAM, and SSFTT but also

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TABLE IV
CLASSIFICATION ACCURACY OF VARIOUS APPROACHES WITH THE KSC DATASET

Class	SSRN	FDSSC	TriCNN	DBDA	HRAM	SSFTT	BoS2T	FTSCN
C1 (20/761)	94.97	97.87	84.30	98.22	**99.86**	74.23	99.75	98.35
C2 (7/243)	86.10	86.02	73.01	81.24	95.76	91.74	**98.16**	96.67
C3 (7/256)	80.90	94.25	45.80	74.71	67.86	66.27	**95.20**	91.32
C4 (7/252)	70.70	63.34	22.71	82.89	79.51	45.60	88.48	**92.28**
C5 (5/161)	70.33	56.78	8.82	95.81	**100.00**	50.93	76.76	90.58
C6 (6/229)	98.40	88.99	52.77	99.10	84.17	80.30	97.10	**100.00**
C7 (3/105)	79.47	94.06	19.18	85.21	96.28	50.25	92.26	**100.00**
C8 (11/431)	96.51	97.79	40.20	98.84	94.91	71.20	**99.42**	99.24
C9 (13/520)	94.45	98.25	37.22	96.57	99.80	65.91	**100.00**	**100.00**
C10 (11/404)	97.69	100.00	39.55	100.00	100.00	53.15	99.99	**100.00**
C11 (11/419)	98.00	97.74	76.88	99.37	99.75	91.59	99.98	**100.00**
C12 (13/503)	99.55	98.60	85.57	99.65	98.36	74.10	**99.99**	99.87
C13 (24/927)	100.00	100.00	98.87	100.00	100.00	90.96	**100.00**	**100.00**
OA/%	92.74 (0.80)	94.34 (1.21)	63.70 (2.39)	95.19 (0.07)	95.16 (0.87)	72.65 (0.62)	97.82 (0.59)	**98.37 (0.41)**
AA/%	89.77 (1.07)	90.28 (2.75)	52.68 (6.11)	93.20 (0.41)	93.57 (0.95)	69.71 (1.59)	95.93 (0.92)	**97.56 (0.45)**
Kappa/%	91.91 (0.90)	93.70 (1.36)	59.50 (2.33)	94.65 (0.08)	94.60 (1.03)	69.38 (1.70)	97.59 (0.66)	**98.18 (0.47)**

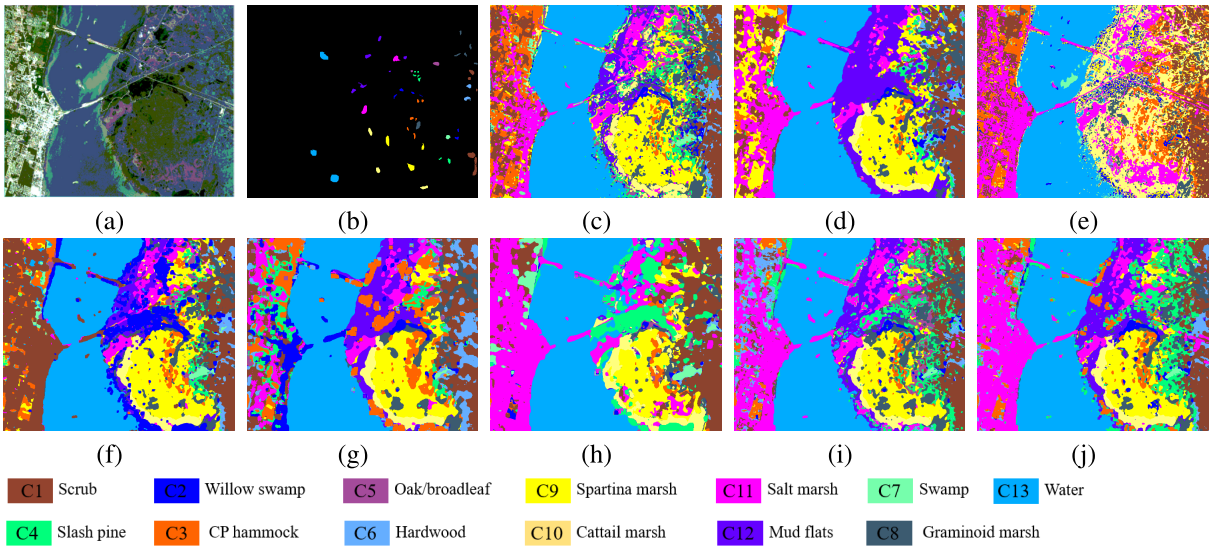


Fig. 10. Classification visual maps obtained by various approaches with the KSC dataset. (a) False-color map. (b) Ground truth. (c) SSRN (92.71%). (d) FDSSC (94.34%). (e) TriCNN (63.70%). (f) DBDA (95.19%). (g) HRAM (95.16%). (h) SSFTT (72.65%). (i) BoS2T (97.82%). (j) FTSCN (98.37%).

have greater smoothness. However, the FTSCN approach performs much better in terms of edge preservation and detail information extraction. This demonstrates the validity and superiority of the proposed FTSCN.

E. Performance With Varying Training Samples

The number of available training samples has a significant impact on the performance of DL-based models. To further investigate FTSCN's performance with different numbers of available samples, experiments with varied proportions of samples are designed to examine the model's sensitivity to sample number differences. The classification results with varying proportions of training samples on UP, IP, KSC, and ZY are displayed in Fig. 12. As illustrated in Fig. 12, the classification accuracy of all DL-based comparison approaches falls as the number of training samples decreases. To make it easier to make visual comparisons of various approaches, the plot of TriCNN is not shown in Fig. 12(c) because the highest accuracy of TriCNN is 72.39%, lower than the lowest value of 81% in the vertical coordinate. Fig. 12(b) adapts the visualization of the resulting plots for clarity. However, even with a lower proportion of training examples in UP, IN, and KSC, FTSCN maintains a higher classification accuracy than all other compared approaches, and the shift in classification results is reasonably smooth as the number of samples declines. In addition, although the classification accuracy of FTSCN varies when the number of samples is relatively small in ZY, it still provides good classification results. This demonstrates that our method is not sensitive to the number of available training samples.

F. Comparison of Parameter Size and Runtime Costs

The number of parameters and running time of the various approaches on different datasets are presented in Table VI. To ensure a fair comparison, we applied an early stopping strategy to all other comparison methods (TriCNN, SSFTT,

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TABLE V
CLASSIFICATION ACCURACY OF VARIOUS APPROACHES WITH THE ZY DATASET

Class	SSRN	FDSSC	TriCNN	DBDA	HRAM	SSFTT	BoS2T	FTSCN
C1 (66/2625)	96.79	98.18	97.01	97.51	97.75	97.50	**98.41**	98.03
C2 (33/1302)	98.03	97.33	96.37	97.49	96.82	**98.21**	98.03	93.09
C3 (87/3442)	98.41	98.73	97.96	98.47	98.36	98.75	98.62	**99.54**
C4 (257/10243)	98.45	98.84	98.30	98.81	98.81	98.97	99.06	**98.94**
C5 (36/1425)	94.88	95.66	89.11	97.34	97.23	95.19	94.61	**98.00**
C6 (38/1484)	96.50	**97.62**	96.50	95.94	95.12	97.34	96.02	97.29
C7 (46/1808)	97.45	97.11	95.68	97.42	97.91	97.17	98.07	**98.30**
C8 (38/1492)	90.37	93.89	91.89	95.97	94.55	90.11	96.12	**98.51**
OA/%	97.14 (0.86)	97.91 (0.46)	96.52 (1.02)	97.96 (0.31)	97.79 (0.67)	97.49 (0.85)	98.10 (0.28)	**98.37 (0.17)**
AA/%	96.36 (0.93)	97.17 (0.80)	95.35 (1.37)	97.37 (0.48)	97.07 (0.73)	96.65 (1.06)	97.37 (0.51)	**97.71 (0.38)**
Kappa/%	96.26 (0.78)	97.26 (0.59)	95.44 (1.25)	97.33 (0.42)	97.10 (0.69)	96.73 (1.23)	97.51 (0.36)	**97.86 (0.21)**

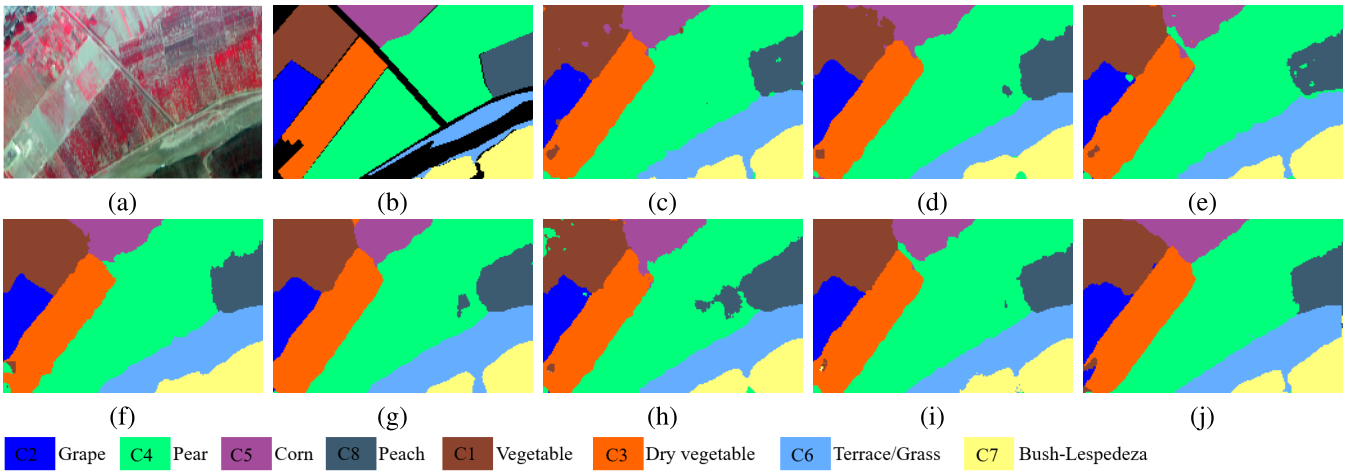


Fig. 11. Classification visual maps obtained by various approaches with the ZY dataset. (a) False-color map. (b) Ground truth. (c) SSRN (97.14%). (d) FDSSC (97.91%). (e) TriCNN (96.52%). (f) DBDA (97.96%). (g) HRAM (97.79%). (h) SSFTT (97.49%). (i) BoS2T (98.10%). (j) FTSCN (98.37%).

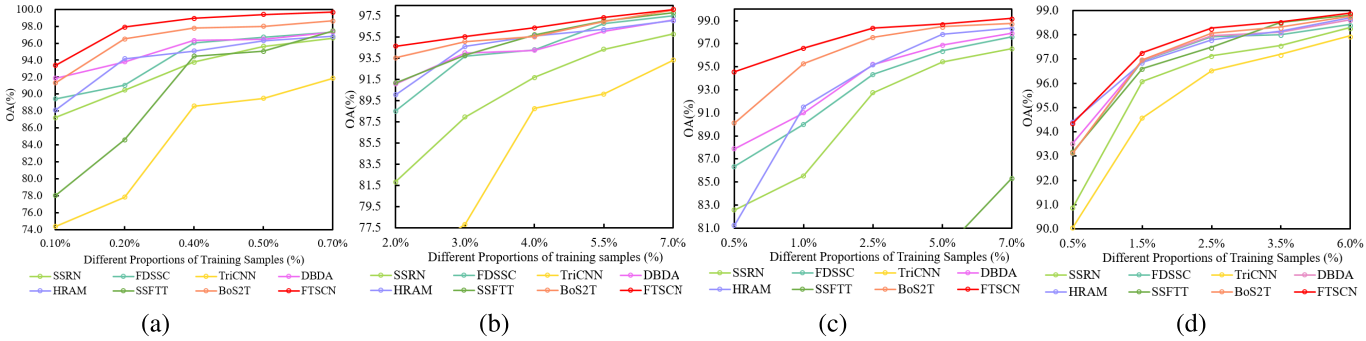


Fig. 12. Classification outcomes obtained by various methods using different proportions of training samples with (a) UP, (b) IP, (c) KSC, and (d) ZY.

and BoS2T) that didn't use it. Compared to other approaches, the FTSCN approach is the least expensive in terms of training time and each epoch, and has a competitive edge in terms of testing time. Notably, the BoS2T method, which is most similar to the proposed FTSCN model framework, not only has larger model parameters than the FTSCN method but also takes more testing time and training time. The reason for this is that the BoS2T approach combines the 3-D-CNN and the BS2T network to extract spatial–spectral feature information, whereas the FTSCN model employs a 2-D-CNN to acquire spatial characteristics and an efficient squeeze-enhanced axial transformer network to extract spectral feature information, greatly reducing the model's complexity. Furthermore, due to the utilization of the new Lion optimizer, the optimal batch size for FSTCN is set to 256, while the optimal batch size for BoS2T is 16, which also contributes to the faster inference speed of the FSTCN model. This is because a larger batch size typically means that more samples can be processed in parallel, allowing them to be simultaneously passed through the forward propagation process of the model, thereby reducing inference time. Similarly, it can be observed that the testing time of the SSRN and HRAM approaches on the KSC dataset is almost the same as that of FTSCN but slower on other datasets. This is because, on the KSC

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dataset, a larger batch size is used to maintain the optimal performance of the model. The FDSSC and DBDA methods require relatively higher training time and testing time, and the time per epoch is also relatively higher compared to FSTCN. This can be attributed to the fact that these 3-D-CNN-based methods not only have deeper network layers and more complex structures but also have a smaller optimal batch size. Smaller batch sizes usually mean that fewer samples can be processed in parallel, leading to a significant increase in inference time. TriCNN employs the 3-D-CNN-based three-branch network (each subnetwork employs a convolutional kernel with various scales); the model uses a large number of FC layers, and its feature dimension is large (up to 512), which greatly increases the parameters and complexity of the whole model (experiments show that the FC layer parameters account for more than 98% of the whole model). It is worth noting that TriCNN takes the longest time for one epoch on UP but is relatively small in terms of testing time. This may be because the higher complexity of the model structure during training necessitates more computations and parameter updates, whereas, during testing, only the forward propagation procedure is required, which is relatively faster. Furthermore, despite increasing its parameters, TriCNN's testing time decreases on other datasets. This is because the batch size grows on other datasets, and higher batch sizes can utilize more parallel processing capability of the computer, speeding up model inference. Moreover, although there is not much difference in testing time between SSFTT and FTSCN on UP, IP, and ZY datasets, it is nearly twice as long for the KSC dataset. This is because, to ensure the optimal performance of the model on KSC, the input patch size is increased to 21×21 (instead of 13×13). This means that the model needs to process more pixels during the forward propagation calculation, resulting in increased computational time.

Even though the ZY dataset contains only 80 spectral bands, and all of its models are not as complex as those on the KSC data (which contains 176 spectral bands). The runtime on the ZY data, however, is longer than on the KSC. This is primarily because the number of samples in ZY (23821) is nearly 4.6 times greater than that in KSC (5211). The larger the dataset, the more samples the model must handle, and thus, the longer the model runtime. It is worth mentioning that, despite the FTSCN model's competitive advantages in terms of runtime and classification performance, the inference speed of FTSCN on the UP, IP, and ZY datasets is not substantially faster than that of SSFTT. This is because SSFTT uses only one 3-D convolutional layer and one 2-D convolutional layer to capture local spectral and spatial features, whereas FTSCN adopts a hierarchical structure based on a dense network to capture layered semantic feature information in order to boost model classification performance, which increases the FTSCN model complexity and, thus, sacrifices some inference speed.

G. Ablation Study

In this section, first, we conducted ablation experiments on the FTSCN approach utilizing various backbone networks to verify the effectiveness of various branching networks. The classification accuracy of various backbone networks with the four considered datasets is presented in Table VII. It clearly shows that the FSTCN model has a higher classification accuracy than the backbone network with simply spectral or spatial branches, demonstrating the effectiveness and significance of the different backbone networks in the feature extraction done by FSTCN.

Subsequently, ablation experiments are conducted for the proposed FTSCN model by selecting various combinations of components to validate the effectiveness of different functional components. Table VIII provides a comparison of the outcome of ablation experiments on various datasets. In this table, NoSima means that the SimAM module is not employed in the construction of the 2-D-CNN spatial feature extraction subnetwork. NoSpam means that the DAU mechanism is not applied in the proposed model. NoSiDAU indicates that neither the SimAM module nor the DAU mechanism is utilized in the spatial branch. NoSeaf indicates that the squeezed enhanced axial transformer module (seaf) is not employed in the spectral branch to simulate the long-range dependence information. NoLion indicates that the model uses the standard Adam instead of the Lion optimizer. NoSeLion indicates that the model neither employs the transformer module to model long-range dependence information nor does it utilize the Lion optimizer. NoSSDL indicates that none of the components in the table is applied.

As seen in Table VIII, the classification accuracy obtained by NoSima is lower than that of the whole FTSCN approach with the four datasets. This is because, without the SimAM module, the proposed approach does not assign more weight to the spatial information-rich neurons in the 2-D-CNN and fails to suppress the redundant information around irrelevant information neurons. Furthermore, since the DAU module with joint spatial and channel attention is applied after spatial characteristic extraction, the key spatial information and feature channels are weighted to help emphasize important features. As a result, NoSpam is less capable than FTSCN in mining relevant feature information. Meanwhile, it can be observed from the outcomes that the categorization performance of NoLion (using the traditional Adam optimizer) is inferior to that using Lion, which suggests the superiority of the Lion optimizer (trained by program search) for improving the model performance. The FTSCN classification approach outperforms the NoSeaf method because the transformer can effectively describe long-range dependencies. Specifically, integrating the squeezed enhanced axial transformer module into the CNN network not only fully utilizes the local feature information but also makes effective use of the long-range feature information, which is beneficial to boost the feature mining ability of the model. Notably, the categorization accuracy of NoSSDL is the lowest of all the datasets. The categorization accuracy of NoSeLion is lower than that of NoSeaf and NoLion, which indicates that not only the squeeze-enhanced axial transformer is effective in improving the feature-mining capability of the model but also the Lion optimizer is efficient in improving the classification performance of the model.

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TABLE VI
NUMBER OF PARAMETERS AND EXECUTION TIMES OF VARIOUS APPROACHES WITH THE FOUR CONSIDERED DATASETS

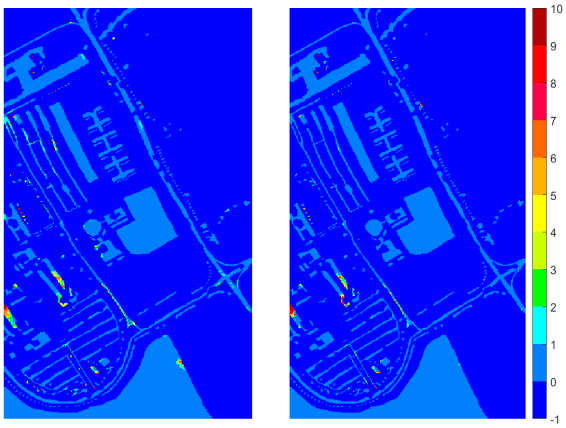
Class		Number of parameters and running times							
		SSRN	FDSSC	TriCNN	DBDA	HRAM	SSFTT	BoS2T	FTSCN
UP	Parameters / M	0.22	0.65	130.22	0.20	0.04	0.15	1.53	0.47
	epoch / s	0.33	0.47	1.02	0.38	0.25	0.16	0.38	0.10
	Training / s	139.41	201.93	122.93	192.21	126.26	59.16	101.43	46.37
	Testing / s	59.92	85.88	33.90	98.53	62.53	56.74	143.85	53.56
IP	Parameters / M	0.39	1.35	305.01	0.42	0.07	0.15	1.74	0.86
	epoch / s	1.40	2.60	2.06	1.68	1.04	0.38	1.50	0.28
	Training	457.76	1027.90	623.66	618.33	515.09	105.80	218.33	85.65
	Testing	19.00	32.12	17.82	39.07	21.72	12.76	40.28	12.13
KSC	Parameters / M	0.33	1.08	490.54	0.34	0.06	0.15	1.66	0.61
	epoch / s	0.37	0.48	1.08	0.45	0.29	0.34	0.42	0.15
	Training / s	143.37	175.73	619.36	212.66	185.59	96.991	113.16	40.20
	Testing / s	8.59	12.90	14.20	15.30	9.57	17.52	17.78	9.27
ZY	Parameters / M	0.18	0.51	238.68	0.16	0.03	0.15	1.49	0.43
	epoch / s	0.98	1.33	2.08	1.21	0.71	0.55	1.15	0.30
	Training / s	378.50	508.80	600.54	449.88	285.33	128.74	283.37	86.71
	Testing / s	30.62	41.81	31.38	55.51	30.68	28.97	70.89	25.55

TABLE VII
CLASSIFICATION ACCURACY OF VARIOUS BACKBONE NETWORKS WITH THE FOUR CONSIDERED DATASETS

Spectral Branch	Spatial Branch	UP	IP	KSC	ZY
✓	×	95.31	94.11	97.13	97.03
×	✓	94.67	90.16	95.32	96.67
✓	✓	**99.00**	**96.38**	**98.37**	**98.37**

TABLE VIII
OUTCOME OF ABLATION EXPERIMENTS ON THE FOUR CONSIDERED DATASETS

Seaf	SimAM	DAU	Lion	UP	IP	KSC	ZY
✓	×	✓	✓	98.79	95.79	97.80	97.96
✓	✓	×	✓	98.83	96.16	97.75	98.05
✓	×	×	✓	98.58	95.60	97.67	97.72
×	✓	✓	✓	98.78	96.18	98.03	97.95
✓	✓	✓	×	98.75	95.61	97.51	98.02
×	✓	✓	×	98.62	95.49	97.30	97.71
×	×	×	×	97.81	95.07	96.80	97.32



(a) (b)

Fig. 13. Misclassification heatmaps of various methods on UP. (a) NoSima. (b) FTSCN. The color bar on the right denotes the number of misclassifications in ten experiments, where −1 denotes the color of the unlabeled pixels.

In addition, to better validate the effectiveness of the SimAM module in FTSCN, visualization experiments have been added. We selected a dataset (e.g., UP) for the visualization experiments, and misclassification heatmaps of various methods (NoSima and FTSCN) on UP are shown in Fig. 13. It is clear from the misclassification heatmap showing the results that FSTCN has significantly fewer misclassified pixels in ten experiments than the NoSima method that does not use the SimAM attention mechanism for the improved CNN. It further demonstrates the effectiveness of using SimAM to improve CNNs in the FTSCN.

V. PRACTICAL HSI APPLICATION

To validate the usefulness and superiority of the FTSCN method in practical scenarios, we performed an experiment with an HSI dataset from a real application in the Dongting area. The data were captured by the GaoFen-5 (GF-5) satellite in 2019. This is the world's first hyperspectral remote sensing satellite capable of observing both land and atmosphere, and contains six payloads for collecting high-resolution remote sensing data in the ultraviolet through long-wave infrared regions [33]. The considered dataset has 456 × 352 pixels, a spatial resolution of 30 m/pixel, and a spectral band coverage range of 400–2500 nm. There are 310 spectral bands altogether after abandoning some noisy bands. This scene has six ground-truth classes with 4816 labeled pixels. In these application data, we randomly pick 1.5% of the sample size as training and validation sets, and the remaining pixels as test samples.

As displayed in Fig. 14, the FTSCN approach outperforms the other comparative algorithms in terms of classification performance, achieving the highest OA score of 98.17%. It is clear that not our proposed technique not only has less classification noise but also exhibits superior edge detail information extraction than the other compared approaches. This is because the FTSCN approach combines CNN with transformer methods, which can effectively leverage both local and long-range information dependencies. Furthermore, the

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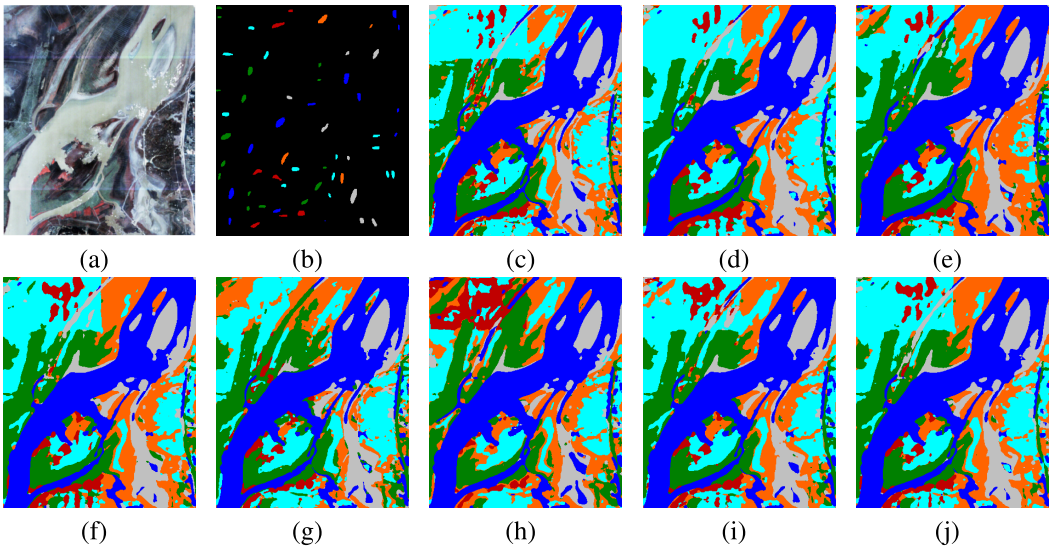


Fig. 14. Classification visual maps obtained by various approaches on the GF-5 dataset over Dongting Lake. (a) False-color map. (b) Ground truth. (c) SSRN (96.55%). (d) FDSSC (97.59%). (e) TriCNN (94.90%). (f) DBDA (97.69%). (g) HRAM (95.71%). (h) SSFTT (94.47%). (i) BoS2T (97.08%). (j) FTSCN (98.17%).

squeeze-enhanced axial transformer captures global semantic characteristics via brief compressed axial attention and augments local detail information using a convolution-based detail enhancement kernel. The results of this experiment also illustrate that our FTSCN can acquire more precise spectral and spatial characteristic information in practical application scenarios.

VI. Conclusion and Future Work

In this study, we propose a new FTSCN method for HSI classification that combines the transformer and a SimAM-based CNN. In order to allocate higher weights to the spatial information neurons and suppress the irrelevant information from surrounding neurons, the SimAM attention module is employed to improve the 2-D-CNN architecture. Thanks to this, the complicated spatial characteristic information of the HSI (stored between each convolutional layer) is retrieved using a hierarchical dense network topology. The DAU mechanism is then utilized to direct the model's attention to discriminative spatial pixels and significant feature maps, so as to boost the model's spatial feature extraction capacity. For the spectral branch, after extracting hierarchical local features from each convolutional layer utilizing a hierarchical 3-D dense network structure, the feature extraction capability of the model is enhanced using a squeeze-enhanced axial transformer combined with a detail-enhanced kernel, where the global long-range dependencies are modeled utilizing the squeezed enhanced axial transformer and the convolution-based detail enhanced kernel is exploited to improve the extraction of local detail features. Finally, a Lion optimizer is utilized instead of the traditional Adam optimizer to enhance the categorization performance of the FTSCN. Our experimental outcomes reveal that the newly proposed FTSCN not only outperforms other state-of-the-art methods in terms of classification performance but also significantly shortens the model training time.

Although building hierarchical dense networks using SimAM-based 2-D-CNNs can effectively extract complex spatial feature information, the performance of using SimAM-based 3-D-CNNs acting on spectral feature extraction at the same time is not satisfactory. This is also our next work that needs to be studied in depth. Meanwhile, considering CNNs based on fixed square kernels fail to perform flexible convolution on the arbitrarily irregular HSI regions, we will also study how to combine the advantages of GCNs and transformers for HSI classification purposes.

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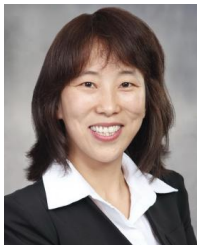
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